SUSTAINABILITY

Closing the Loop: Structural Steel Reuse in Hyperscale Datacenter Construction

A Scalable Blueprint from Microsoft's Newport Imperial Park Datacenter

EUROPE Regional Community

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Agenda

1. Microsoft’s Sustainability Commitments
2. Project Overview
3. Project Phases
    a. Planning & Stakeholder Engagement
    b. Design Phase
    c. Deconstruction Phase
    d. Construction Phase
4. Key Learnings
5. Call to Action

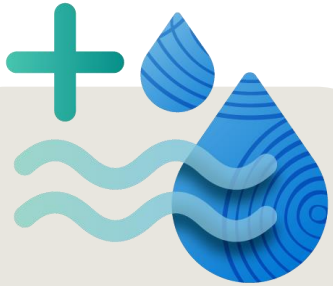


Microsoft’s Sustainability Goals



Carbon negative

Our sustainability journey starts with reducing our carbon emissions as much as possible, including via matching our energy consumption with carbon-free energy, and removing any emissions that remain.



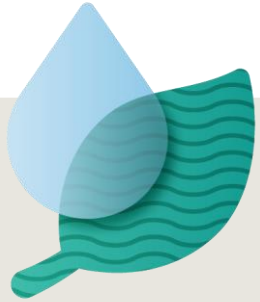
Water positive

At Microsoft, net water positive means reducing water consumption across our global operations, replenishing more water than we use, providing people across the globe with access to water and sanitation services, driving innovation, and engaging in water policy.



Zero waste

We’re taking an increasingly circular approach to materials management. Our strategy goes beyond waste diversion as we work across our value chain, beginning with design and material selection, to reduce waste.

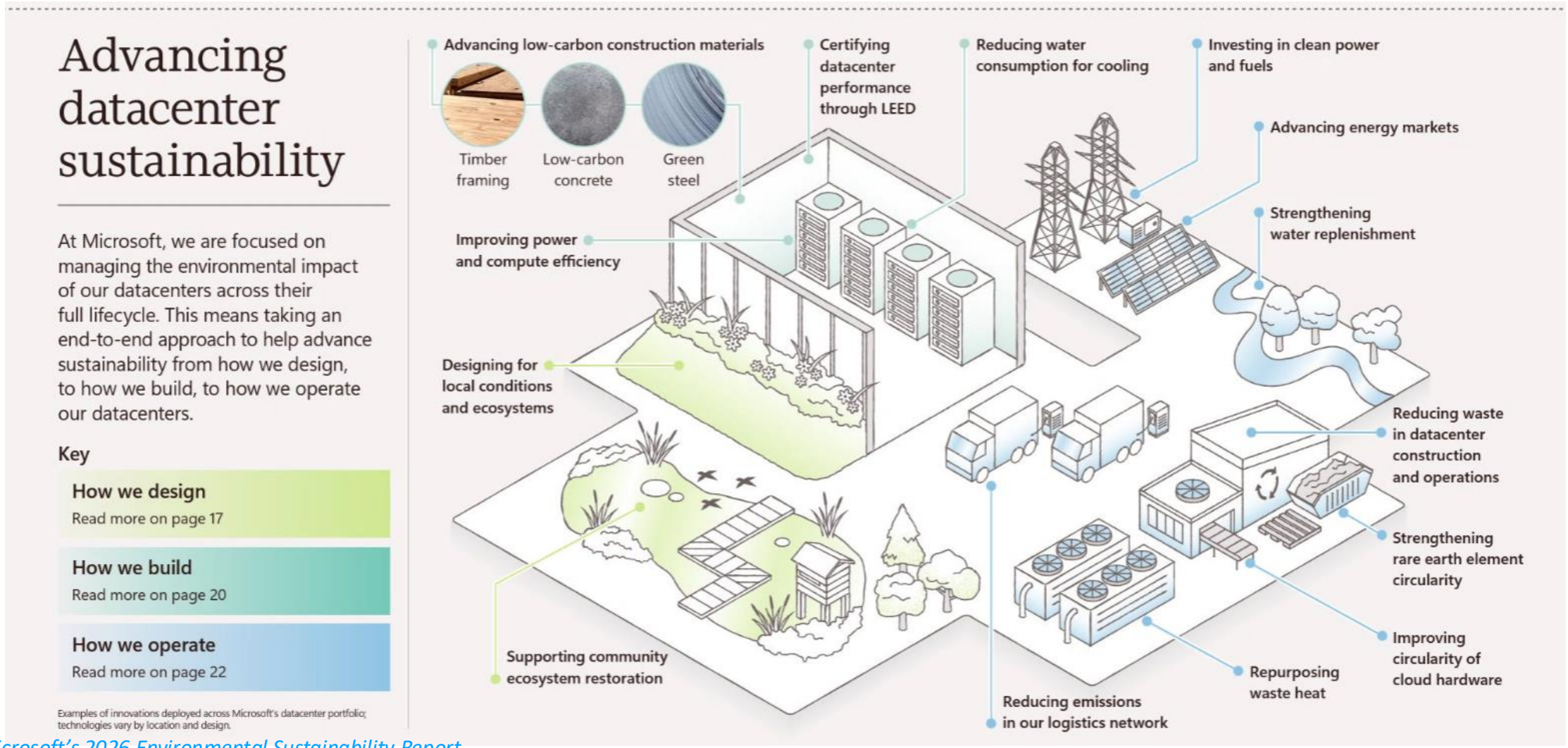


Protect ecosystems

We achieved our goal of protecting more land than we use in 2022.

Protecting more land than we use means going beyond our own operations by actively protecting the environmental health of the communities that host our datacenter operations and where our employees live and work.

End-to-End Sustainability Approach at Microsoft



Source: [Microsoft's 2026 Environmental Sustainability Report]

Why Circularity in Datacenter Construction

- Microsoft's sustainability commitments require lower-impact construction that can keep pace with global datacenter growth. Structural steel is carbon-intensive to produce, yet durable, testable, and well suited to direct reuse when provenance and acceptance are managed.

Newport Imperial Park Datacenter moves beyond diversion and recycling toward adaptive reuse: recovering structural members from an existing facility, certifying them, and reintegrating them into a new build.

STARTING FROM STRENGTH

+ 90%

In FY25, Microsoft and our partners diverted over 90% of our construction and demolition waste from landfills and incinerators.

4 phases

- Planning & stakeholder engagement
- Design Phase
- Deconstruction Phase
- Construction Phase

1st

End-to-end structural steel reuse pilot in Microsoft datacenter construction



NEWPORT, SOUTH WALES

Project Overview

Newport Imperial Park Datacenter: A circular construction pilot

- The former Quinn Radiator complex provided a substantial steel-framed donor structure. Microsoft and delivery partners coordinated surveying and modeling, tagging and selective removal, testing and certification, staging, and reintegration
- Early surveys established a ~13% reclaimed-steel target. The project ultimately reused 249 tonnes, roughly 10% of the structural frame, and supplemented the balance with high-recycled-content steel.

Legacy steel-framed donor structure

249 tonnes reclaimed

No construction-phase cost or schedule impact

The Impact at Newport Imperial Park Datacenter

249 tons

structural steel reclaimed

Approximately 10% of structural steel reused from the former facility

76%

recycled-content steel

High-recycled-content steel across the structural package

~520 tCO_2e

saved through reuse

Savings attributable to reclaimed steel versus new production

~4,400 tCO_2e

total steel strategy

Combined embodied-carbon savings from reclaimed and recycled steel

+ 80%

waste diverted

Construction and deconstruction program outcome

+ 90%

deconstruction diversion

Site waste diversion rate during deconstruction

Delivered without construction-phase cost overruns or schedule impacts because steel reuse was incorporated at tender stage.

Stakeholder Coordination Across the Delivery Model

Stakeholder coordination is the core enabler: roles, responsibilities, and structural acceptance criteria were defined at tender stage, reducing ambiguity and enabling parallel work across disciplines.

1 Owner / Client Sponsor

Sets circularity objectives, convenes stakeholders, and coordinated delivery risk across the program.

2 Engineering & Reuse Specialist

Designed and validated reuse, defined the roadmap, and governed structural acceptance.

3 Deconstruction Contractor

Leads selective deconstruction, member identification, controlled handling, and traceability.

4 Construction Contractor

Manages testing, acceptance, fabrication alignment, and erection sequencing on site.

DELIVERY MODEL WORKFLOW

Align Intent & Set Targets → **Recover & Tag Materials** → **Test, Certify & Accept** → **Integrate & Build**

The Workflow: From Old Frame to New Build

01 Design & Early Intent

Set reuse targets and design parameters before structural systems are fixed.

02 Selective Deconstruction

Recover high-value members through controlled sequencing and handling.

03 Testing & Certification

Use NDT, destructive testing, documentation, and beam-level traceability.

04 Construction Integration

Review evidence, screen, model, fabricate, and erect accepted members.

Design Phase

Early intent preserves optionality

01 Structured roadmap

Consulting Engineer's framework defined feasibility, testing, certification, and documentation early.

02 Decide before Stage 3

Beyond RIBA Stage 3, structural changes introduce greater schedule and program risk.

03 Class B reuse case

Missing original mill certificates required strengthened testing and traceability.

04 Design for variable stock

Grouping, cut-length tolerances, and connection strategies enabled practical integration.

Deconstruction Phase

Turning Reclaimed members into Trusted Inputs

Beam-level traceability linked each member to its original location, condition assessment, test evidence, acceptance status, and final installation.

1. **Identify & classify**
2. **Inspect & assess**
3. **Test & document**
4. **Certify & accept**

SCI P427 • BS EN 1090 • CC2 assurance



Deconstruction Phase cont.

Recovery as Managed Production

- ✓ Phased sequence: soft strip, enabling works, structural deconstruction, slab removal
- ✓ Cranes instead of excavators for reuse-critical members
- ✓ Temporary bracing and controlled lifting to prevent distortion
- ✓ Minimal handling after removal to protect member quality

+ 90% deconstruction waste diversion





Construction Phase

Reclaimed Steel as a Standard Input

01 Documentation review
Verify test and inspection evidence for each member.

02 Screening & acceptance
Reject damaged or non-compliant material before integration.

03 Model integration
Add approved beams and columns to the structural model.

04 Fabrication & erection
Erect reclaimed and recycled-content steel in one coordinated frame.

Circularity Beyond Steel

Environmental and Social Value

Water Stewardship

3 attenuation ponds

1 subsurface rainwater tank

Designed to reduce potable-water demand for cooling under normal operations.

Biodiversity Uplift

+ 260% habitat units

+ 620% hedgerow units

Anticipated planning gains through retained trees and native planting.

Community & Skills

28 apprentices / 27 graduates

13 student placements

69% of workforce lived within 40 km of the site.

LEED Gold-aligned practices • FSC/PEFC timber • HVO fuel • renewable and EV infrastructure • zero environmental incidents

Programmatic Learnings

Six Transferable Principles

1 Shared understanding

Align sustainability goals and acceptance expectations early.

2 Early intent

Preserve optionality before design and procurement pathways are fixed.

3 Cross-disciplinary collaboration

Manage recovery, testing, design, and fabrication as one system.

4 Traceability

Use component-level evidence to reduce risk and enable acceptance.

5 Regional context

Adapt to regulation, testing capacity, and secondary-material markets.

6 Standardized workflows

Repeatable testing, approval, and handover improve scalability.

Call to Action

From one-site proof point to portfolio capability

Newport Imperial Park Datacenter proves the technical and delivery model. Scaling requires the market infrastructure and governance that make reuse routine rather than exceptional.

01 Reclaimed supply chains

Build donor-building pipelines, storage, and reuse channels.

02 Testing capabilities

Standardize NDT, destructive testing, and certification pathways.

03 Procurement frameworks

Embed reuse criteria into tenders, contracts, and contractor evaluation.

04 Circularity Ecosystem

Create a shared standards-aligned framework that accelerates industry adoption.

Regional feasibility varies

EMEA: recovery mandates and pre-deconstruction audits • APAC: market- and jurisdiction-specific sorting/recycling pathways • Americas: state, provincial, municipal, and market capacity shape outcomes

Thank You!

